Coordinator & Worker Swarm Execution

How Complex High-Level Tasks Are Broken Down and Executed in Parallel

Dynamic Hierarchical Task Decomposition

When a user assigns a complex project, the Coordinator uses a structured planning prompt to analyze dependencies, formulate execution branches, and launch parallel sub-agents via the `spawn_agent` tool.

Sample Task Decomposition Tree

Root Coordinator: "Comprehensive Due Diligence on European AI FinTechs"

Branch 1 (Registry Agent): Scrapes Companies House & Handelsregister for corporate registration & filings.

Branch 2 (OSINT Agent): Harvests C-level leadership, LinkedIn profiles, and verified work emails.

Branch 3 (Tech Stack Agent): Fingerprints homepage HTML for 40+ technology signatures (AWS, Stripe, Next.js).

Synthesis Node (Analyst & Writer): Merges all 3 branches, reconciles conflicts, and formats final executive PDF report.

RECURSIVE DEPTH LIMITS & GUARDRAILS

To prevent runaway sub-agent spawning loops, Fathom enforces strict depth limits (default max depth: 2). Coordinators can spawn workers, but workers cannot spawn infinite child trees without explicit permission.

TOKIO JOINSET MULTI-THREADING

All spawned workers run as concurrent tasks inside a Tokio `JoinSet`. If one branch encounters a slow network request or rate limit, sibling branches continue executing at full speed.

TOKEN BUDGET FAIR-SHARE SCHEDULING

The coordinator assigns token quotas to each sub-agent branch (e.g. max 16k tokens per worker). When a worker approaches 80% of its budget, it enters Economy Mode, truncating verbose tool outputs to prevent waste.

Fault-Tolerant Merging: If one sub-agent fails due to an anti-bot block, the coordinator gracefully incorporates partial findings from other branches without failing the overall run.